

Name: Charlie Pyle

1. Use the definition of an even integer to explain why 180 is even.

Solution: An integer n is even provided that there exists an integer a such that $n = 2a$.

We have $180 = 2(90)$, and 90 is an integer. Therefore, by the definition of an even integer, 180 is even.

2. Identify the hypothesis and conclusion for each of the following conditional statements. One of these statements is not written in “if... then...” format. First you must rewrite it in that format.

(a) If x is a negative real number, then x^3 is a negative real number.

(b) All cats are cute.

Solution: Part 1:

Hypothesis: x is a negative real number.

Conclusion: x^3 is a negative real number.

Part 2:

The statement “All cats are cute” is not written in if-then format. Rewritten in if-then format, it reads: “If an animal is a cat, then the animal is cute.”

Hypothesis: An animal is a cat.

Conclusion: The animal is cute.

3. Consider the following statement:

Statement: If a and b are odd integers and n is an even integer, then $a^2n - b^2 + n$ is an odd integer.

(a) Give two examples supporting this statement.

(b) Construct a know-show table outlining an argument you would use to prove the statement above.

(c) Write a complete proof for the statement above.

Solution: Part 1: Two supporting examples

Let $a = 1$, $b = 1$, and $n = 2$. All three integers satisfy the hypotheses: a and b are odd, and n is even. We compute

$$a^2n - b^2 + n = 1^2(2) - 1^2 + 2 = 2 - 1 + 2 = 3,$$

and 3 is an odd integer, as claimed.

Let $a = 3$, $b = 5$, and $n = 4$. Again a and b are odd, and n is even. We compute

$$a^2n - b^2 + n = 3^2(4) - 5^2 + 4 = 36 - 25 + 4 = 15,$$

and 15 is an odd integer, as claimed.

Part 2: Know-show table

Step	Know	Reason
P	a and b are odd integers, and n is an even integer.	Hypothesis
$P1$	There exist integers k and m such that $a = 2k + 1$ and $b = 2m + 1$; there exists an integer j such that $n = 2j$.	Definition of odd integer; definition of even integer
$P2$	$a^2n - b^2 + n = 2(4k^2j + 4kj + 2j - 2m^2 - 2m - 1) + 1$.	Algebra: substitute $P1$ into $a^2n - b^2 + n$ and simplify
$P3$	$p = 4k^2j + 4kj + 2j - 2m^2 - 2m - 1$ is an integer.	Closure properties of addition and multiplication for integers

Step	Show	Reason
$Q1$	There exists an integer p such that $a^2n - b^2 + n = 2p + 1$.	Matches $P3$, since p is an integer
Q	$a^2n - b^2 + n$ is an odd integer.	Definition of odd integer

Part 3: Complete proof

Theorem. If a and b are odd integers and n is an even integer, then $a^2n - b^2 + n$ is an odd integer.

Proof. We assume that a and b are odd integers and that n is an even integer, and we will prove that $a^2n - b^2 + n$ is an odd integer.

Since a is an odd integer, there exists an integer k such that

$$a = 2k + 1.$$

Since b is an odd integer, there exists an integer m such that

$$b = 2m + 1.$$

Since n is an even integer, there exists an integer j such that

$$n = 2j.$$

Using algebra, we obtain

$$\begin{aligned}
a^2n - b^2 + n &= (2k + 1)^2(2j) - (2m + 1)^2 + 2j \\
&= (4k^2 + 4k + 1)(2j) - (4m^2 + 4m + 1) + 2j \\
&= 8k^2j + 8kj + 2j - 4m^2 - 4m - 1 + 2j \\
&= 8k^2j + 8kj + 4j - 4m^2 - 4m - 1 \\
&= 2(4k^2j + 4kj + 2j - 2m^2 - 2m - 1) + 1.
\end{aligned}$$

An integer p can be defined by

$$p = 4k^2j + 4kj + 2j - 2m^2 - 2m - 1.$$

Since k , m , and j are integers, p is an integer as well, by the closure properties of addition and multiplication for integers. We have shown that

$$a^2n - b^2 + n = 2p + 1,$$

where p is an integer. By the definition of an odd integer, we conclude that $a^2n - b^2 + n$ is an odd integer. This completes the proof. ■